

A Web-Based Machine Learning Application for Employee Attrition Prediction

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Abstract—Employee attrition is a critical issue faced by organizations, a major challenge faced by organizations worldwide, directly impacting workforce stability, productivity, and operational costs. Traditional methods of analyzing attrition often rely on descriptive statistics, which may not fully capture complex interactions among demographic, behavioral, and organizational factors. This paper proposes a machine learning approach to predict employee attrition using a Random Forest Classifier with both numerical and categorical features, presents a balanced approach combining machine learning and web-based deployment for predicting employee attrition using the IBM HR Analytics dataset. The model achieved an accuracy of 87.41%, indicating its effectiveness in identifying employees who are likely to leave or remain in the organization. Feature importance analysis revealed that total working years, age, and overtime status significantly influenced attrition. To enhance accessibility and interactivity, the trained model was deployed through a Streamlit web application, allowing user to input attributes and obtain real-time predictions. The results reveal that positions such as age between 30-40, no overtime (30%), and marital status is divorced (25%) are the most influential features linked to attrition. This integration of predictive analytics and interactive visualization provides a practical tool for human resource decision making and proactive workforce management. The results demonstrate that machine learning can provide actionable insights to support resource strategies in retaining employees and reducing turnover risks.

Index Terms—employee attrition, prediction, machine learning, web application, streamlit, random forest

I. INTRODUCTION

Employee attrition, which is when employees leave an organization, has become a critical issue for companies in this modern era. Because the competition for workers is higher now, and the values of the new generation are different, it is more difficult to manage the stability of the workforce. Because of this, the attrition rate is now an important measurement to see the health and sustainability of a company [1].

High levels of attrition lead to substantial organizational costs, including increased recruitment and training expenses, productivity disruption, and the loss of institutional knowledge. Moreover, attrition negatively affects employee morale and team cohesion. Hence, identifying the underlying factors that contribute to employee attrition has become a top priority

for organizations aiming to enhance retention through data-driven decision-making [2].

Within the field of Human Resource (HR) Analytics, machine learning (ML) has emerged as a powerful approach for modeling and predicting employee attrition. Despite the advancements, challenges remain in balancing model performance with fairness and generalizability. Data imbalance, sampling bias, and overfitting to specific organizational contexts can limit the reliability of predictive systems [3]. Furthermore, privacy and ethical considerations in handling employee data require careful attention, especially when deploying predictive tools in real-world HR environments. Addressing these challenges requires the development of systems that combine technical accuracy, ethical design, and practical usability.

However, most existing studies primarily focus on the machine learning modeling aspect, with limited emphasis on usability or deployment. The practical challenge remains: how can HR managers easily access and interpret predictive analytics without technical expertise? Developing an interactive web-based application can bridge this gap by allowing users to input employee characteristics, obtain real-time predictions, and visualize attrition probabilities directly.

The integration of ML models into user-friendly platforms has been a growing trend in predictive HR analytics. Web-based systems, for example the one made with Streamlit, help HR professionals use predictive models even if they have no skill in programming. This system can give prediction and visualization in real time, making complex analytics can be accessed from an easy to use interface. This is important to make machine learning research become a practical application to support company decision-making. Therefore, this study aims to build a system that is accurate and also easy to use. With this system, user can explore the dynamics of attrition, test some scenarios, and make good decisions with a user-friendly interface.

II. RELATED WORKS

The problem to predict and understand employee turnover is a broad focus in many papers. Researchers are using more computer and statistical methods to help human resource management. Using machine learning is now a popular way to find which employees might leave the company.

A systematic review from Talebi [2] gives a big picture of machine learning for turnover prediction. This paper looks at many studies and shows that algorithms like Random Forest, Support Vector Machines (SVM), Logistic Regression, and others are popular and effective. The review suggests that while these models are powerful, their true value comes from data that reflects the real experiences of employees. The most significant insights emerge when we consider key aspects of an employee's work life, such as their sense of fulfillment (job satisfaction), financial well-being (salary), and their personal journey with the company (tenure).

Building on this, Chen's [4] work with the IBM dataset illustrates how statistical tools can be used to listen to an organization's story. By visualizing the connections between compensation, work-life balance, and an employee's choice to leave, the study highlights the importance of understanding the unique culture and challenges within a specific workplace to foster a more supportive environment.

In this field, researchers want to find tools that can understand why people make decisions at work, because it is very complex. Many models were tried, but Random Forest is often the best one. It is good because it can understand the many different reasons that affect a person's job [5]–[10]. A study from a company in Italy also showed that Random Forest was the best to understand their employee data [11]. Because of this, we chose Random Forest for our study. It is not just because it is good at predicting, but because it can also understand that an employee's experience has many sides.

However, a prediction alone is not enough. To truly support employees, HR leaders need to understand the stories behind the data. Several studies emphasize that the "why" is as important as the "what" [11]. Tools that can explain a model's reasoning, by highlighting factors like job role, work-life balance, or overtime, transform a simple prediction into a conversation starter. This transparency is crucial for developing compassionate and effective policies that address the root causes of employee dissatisfaction, rather than just reacting to the symptoms.

To add to this idea, other research also looks at how the company environment can change an employee's experience. For example, the research from Saufi et al. [12] reminds us that things like fair HR rules and a good company name are very connected to an employee's feeling of belonging. They found that work-life balance is a very important connection for this. This shows us a big picture view: an employee's decision to stay or leave is not made for just one reason. It is a very personal choice that is affected by their whole life at work and outside of work.

III. METHODOLOGY

To connect the information from data with real action for people, this study uses a clear method. Our way is to make not just a predictive model, but also a tool that is easy to use for the HR people. The process has four steps: first, understanding the stories in the data, second, preparing the data carefully, third, making a good predictive model with Random Forest, and last,

building a web application that is easy to use. Our goal is to make a process that is clear and can be followed again, so the final tool can help HR professionals to better understand and support their colleagues.

A. Dataset Description

The starting point for our exploration is the publicly available "IBM HR Analytics Employee Attrition & Performance" dataset. This rich collection of information provides a window into the professional lives of 1,470 individuals. Each profile is described by 35 distinct attributes, which together create a mosaic of their experiences. These attributes encompass a wide range of information, including:

- Demographic data: age, gender, marital status, and education.
- Job-related information: department, job role, job level, business travel, and overtime.
- Employee satisfaction metrics: job satisfaction, performance rating, and work-life balance.

The target variable for our prediction task is Attrition, a binary feature indicating whether an employee has left the company ("Yes") or remained ("No").

B. Data Preprocessing

Before we trained the model, we prepared the raw data first. First, we changed the categorical features like Department, Gender, and Job Role into numbers. We used one-hot encoding for this. This method makes new columns with 0 and 1 for each category. This helps the algorithm to understand the data without thinking one category is more important than another. After that, we split the data into a training set and a testing set. We used an 80/20 split. So, 80% of the data was for training the model, and the other 20% was saved for testing. We used this 20% to see how good the model is with new data.

C. Predictive Modeling with Random Forest

For the prediction task, a Random Forest classifier was selected as the primary algorithm. Random Forest is an ensemble learning method that operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes of the individual trees. This model was chosen for its robustness against overfitting, its ability to handle both numerical and categorical data effectively, and its high performance in classification tasks [13]. The Random Forest model was trained on the 80% training subset. After training, the model's predictive performance was evaluated on the 20% testing subset. The primary metric used for the evaluation was accuracy, which measures the proportion of total predictions that were correct.

D. Web-Based Visualization and Prediction System

To make the model's power become a useful tool that is easy to use, we made an interactive web application using Streamlit. Streamlit is an open-source library in Python. This application has an interface that is easy to understand. Here,

a user like an HR manager can put the values for employee features using sliders and dropdown menus.

After the user submits the input, the application will give this data to the Random Forest model directly. The model will process the data and give a prediction that is clear and easy to understand: if the employee will leave ("Attrition: Yes") or stay ("Attrition: No"). This application is a proof-of-concept. It shows how we can use machine learning models in a real way to help HR management make decisions with data.

IV. RESULTS

A. Model Performance and Evaluation

The Random Forest classifier was trained and evaluated, and its performance on the unseen test data is summarized by the confusion matrix. The model achieved an overall accuracy of 87.41%, calculated from the predictions on the test set. To understand the model's classification performance in detail, a confusion matrix was generated, as shown in Figure 1. This matrix provides a clear breakdown of correct and incorrect predictions for both classes (Stay vs. Leave).

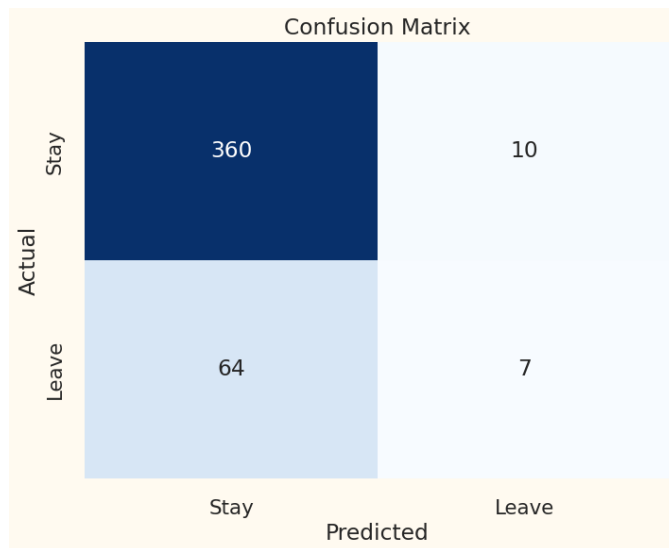


Fig. 1. Confusion matrix of the Random Forest model.

The matrix reveals the following:

- True Negatives (TN): 360 – The model correctly predicted that 360 employees would stay.
- True Positives (TP): 7 – The model correctly predicted that 7 employees would leave.
- False Positives (FP): 10 – The model incorrectly predicted that 10 employees would leave when they actually stayed (Type I error).
- False Negatives (FN): 64 – The model incorrectly predicted that 64 employees would stay when they actually left (Type II error).

Although the model is very effective at identifying employees who will stay, the higher number of False Negatives indicates a challenge in correctly identifying all the employees who are likely to leave. This is a common issue in imbalanced

datasets where the number of employees leaving is much smaller than those staying.

B. Interactive Prediction Application

To demonstrate the practical utility of the predictive model, an interactive web application was developed using Streamlit. This tool allows the user to leverage the model's capabilities by inputting the data and receiving an instant prediction of their attrition risk.

Beyond its predictive capabilities, the application functions as an interactive dashboard. It includes several collapsible sections that provide users with visual summaries of the underlying dataset, allowing for a deeper understanding of attrition trends. These sections include:

- Overall statistics: Provides a high-level view of the dataset.
- Statistics by personal data: Breaks down attrition by demographic factors.
- Statistics by employee detail: Analyzes attrition based on employment-related attributes.
- Statistics by job: Segments attrition data by job characteristics.

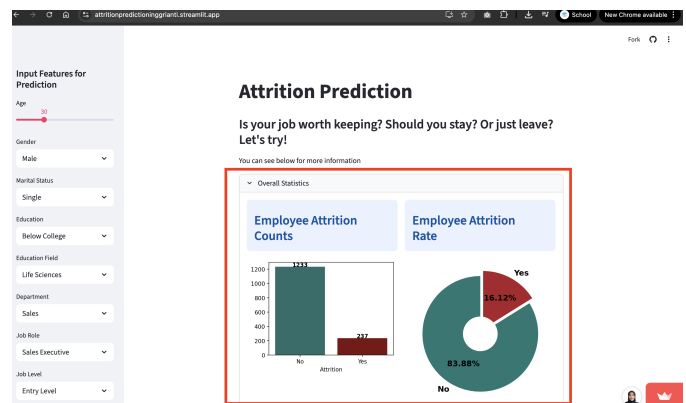


Fig. 2. Overall statistics view in the application.

In Figure 2, for example, the overall statistics section gives a quick look at the employee attrition situation in the dataset. The bar chart on the left side shows the real numbers: 1,233 employees did not leave, and 237 employees left. The pie chart on the right side shows this in percent, so the total attrition rate is 16.12%. This first look is very important to understand how big the attrition problem is and also to see that the data is not balanced. In the other sections, users can see more detail and explore the attrition rates for different groups of employees. This makes the tool very useful for exploring the data.

Figure 3 shows the big picture of the application interface. As we can see in the highlight, the left side is for the "Input Features for Prediction." Here, a user can change the employee data like Age, Gender, Education, Job Role, and more. The main part shows the "Model Accuracy," so we can be clear about how good the model is.

The main function of the application is to give real-time predictions from these inputs. A special feature is that it can

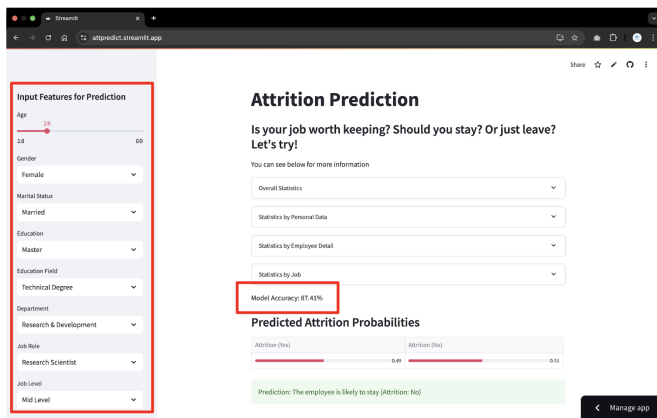


Fig. 3. Overview of the attrition prediction application.

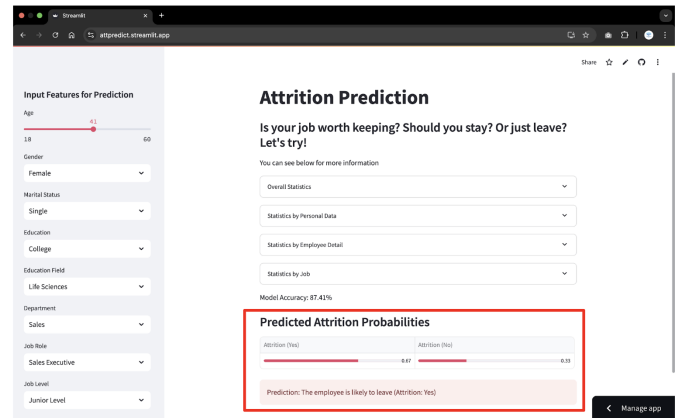


Fig. 5. Application showing a "leave" prediction.

show not only the final prediction, but also the probabilities. This gives a more detailed view of how confident the model is.

receive immediate feedback, the tool empowers proactive and data-informed decision-making.

V. CONCLUSION

This paper has successfully shown how to develop and use a machine learning system to predict employee turnover. By using the IBM HR Analytics dataset and a Random Forest classifier, we built a strong model with 87.41% accuracy to find employees who might leave their job.

The main contribution of this research is not only making the model, but also making the practical application. We developed an interactive web application with Streamlit. This application changes the model's power into a tool that is easy for HR professionals to use. This application shows how companies can use data science to support their strategies for keeping employees.

While there are limits because we used a fictional dataset, but this study gives a complete method that can be repeated. For the future, this method can be used on real company data and add more features to the application to explain the predictions better. This will give more useful information. In the end, this research shows the big potential of machine learning to make human resource management become smarter, more responsive, and more effective.

REFERENCES

- [1] J. C. M. Bustillo, "Optimization-based techniques prediction model in determining employee turnover," *Procedia Computer Science*, pp. 440–449, 2025.
- [2] H. Talebi, A. Khatibi Bardsiri, and V. K. Bardsiri, "Machine learning approaches for predicting employee turnover: A systematic review," *Engineering Reports* 7, no. 8, 2025.
- [3] N. B. Yahia, J. Hlel and R. Colomo-Palacios, "From big data to deep data to support people analytics for employee attrition prediction," *IEEE Access*, vol. 9, pp. 60447–60458, 2021.
- [4] B. Chen, "Factors of employee attrition: A logistic regression approach," in *International Conference on Management Research and Economic Development*, 2023.
- [5] S. S. Alduayj and K. Rajpoot, "Predicting employee attrition using machine learning," in *Proc. Int. Conf. Innov. Inf. Technol. (IIT)*, AI Ain, UAE, pp. 93–98, November 2018.
- [6] M. Ganesh V. Aishwaryalakshmi, S. Akshaya, and K. Abinaya, "Predicting employee attrition using machine learning," *Int. J. Sci. Res. Comput. Sci., Eng. Inf. Technol.*, vol. 3, no. 3, pp. 145–149, 2018.

Figure 4 is an example when the prediction for an employee is to stay. In the 'Predicted Attrition Probabilities' section, it shows the probability is 0.51 for 'Attrition (No)' and 0.49 for 'Attrition (Yes)'. The final prediction has a green box, so it is easy to read. Different from that, Figure 5 shows a case with high risk. Here, the model predicts the employee will probably leave, with a probability of 0.67 for 'Attrition (Yes)' and 0.33 for 'Attrition (No)'. The result has a red box to give a clear warning. This information can help HR to prioritize their actions and focus on employees who have the highest risk.

The making of the Streamlit application is an important part of this work. Many studies only focus on making and testing the model [3], but our research connects the model with a real application. This interactive tool can change the difficult output from the machine learning model to become a useful system for HR to make decisions. This is the same with the trend to make AI more easy to access and to understand. By allowing the user to simulate different employee scenarios and

- [7] Y. Zhao, M. K. Hryniewicki, F. Cheng, B. Fu, and X. Zhu, "Employee turnover prediction with machine learning: A reliable approach," in *Intelligent Systems and Applications*, vol. 869, K. Arai, S. Kapoor, and R. Bhatia, Eds. Cham, Switzerland: Springer, pp. 737–758, 2019.
- [8] X. Gao, J. Wen, and C. Zhang, "An improved random forest algorithm for predicting employee turnover," *Math. Problems Eng.*, vol. 2019, pp. 1–12, April 2019.
- [9] S. Karande and L. Shyamala, "Prediction of employee turnover using ensemble learning," in *Ambient Communications and Computer Systems*, vol. 904, Y.-C. Hu, S. Tiwari, K. K. Mishra, and M. C. Trivedi, Eds. Singapore: Springer, pp. 319–327, 2019.
- [10] F. Fallucchi, M. Coladangelo, R. Giuliano, and E. W. De Luca, "Predicting employee attrition using machine learning techniques," *Computers*, vol. 9, no. 4, p. 86, November 2020.
- [11] S. M. Varkiani, F. Pattarin, T. Fabbri, G. Fantoni, "Predicting employee attrition and explaining its determinants," *Expert Systems with Applications*, vol. 272, 2025.
- [12] R. A. Saufi, S. Aidara, N. C. Che Nawi, R. Permarupan, N. R. B. Zainol, and A. S. Kakar, "Turnover intention and its antecedents: The mediating role of work-life balance and the moderating role of job opportunity," *Frontiers in Psychology*, vol. 14, 2023.
- [13] A. Liaw and M. Wiener, "Classification and regression by randomForest," *R News*, vol. 2, pp. 18–22, December 2002.